

CHE 312 HW #6

Wednesday, October 15, 2025

11:33 PM

9.3

$$Ra = \frac{g \beta \Delta T L^3}{\nu \alpha}$$

where $\alpha = \frac{k}{\rho c_p}$, $\nu = \frac{\mu}{\rho}$
 $\beta = 1/T$ for ideal gases

1. Air (400K)

$$\mu = 2.29 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

$$k = 0.036 \text{ W/m}\cdot\text{K}$$

$$\rho = 0.882 \text{ kg/m}^3$$

$$c_p = 1007 \text{ J/kg}\cdot\text{K}$$

$$\nu = \frac{\mu}{\rho} = 2.60 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\alpha = \frac{0.036}{(1007)(0.882)} = 4.05 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Ra = \frac{(9.81)(0.0025)(30)(0.01)^3}{(2.60 \times 10^{-5})(4.05 \times 10^{-5})}$$

$$Ra = 699$$

2. Helium (400K)

$$\mu = 3.0 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

$$k = 0.15 \text{ W/m}\cdot\text{K}$$

$$\rho = 0.122 \text{ kg/m}^3$$

$$c_p = 5196 \text{ J/kg}\cdot\text{K}$$

$$\nu = \frac{3.0 \times 10^{-5}}{0.122} = 2.46 \times 10^{-4} \text{ m}^2/\text{s}$$

$$\alpha = 5.68 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Ra = \frac{(9.81)(0.0025)(30)(0.01)^3}{(2.46 \times 10^{-4})(5.68 \times 10^{-5})}$$

$$Ra = 52.7$$

3. Glycerin (285 K)

$$\mu = 1.50 \text{ Pa}\cdot\text{s}$$

$$k = 0.285 \text{ W/m}\cdot\text{K}$$

$$\rho = 1260 \text{ kg/m}^3$$

$$c_p = 2410 \text{ J/kg}\cdot\text{K}$$

$$\nu = 1.19 \times 10^{-3} \text{ m}^2/\text{s}$$

$$\alpha = 9.39 \times 10^{-8} \text{ m}^2/\text{s}$$

$$Ra = \frac{(9.81)(6 \times 10^{-4})(30)(0.01)^3}{(1.19 \times 10^{-3})(9.39 \times 10^{-8})}$$

$$Ra = 1580$$

4. Water (310K)

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$$\mu = 6.91 \times 10^{-4} \text{ Pa}\cdot\text{s} \quad K = 0.63 \text{ W/m}\cdot\text{K}$$

$$\rho = 993.3 \text{ kg/m}^3 \quad C_p = 4197 \text{ J/kg}\cdot\text{K}$$

$$\nu = 6.96 \times 10^{-7} \text{ m}^2/\text{s} \quad \alpha = 1.51 \times 10^{-7} \text{ m}^2/\text{s}$$

$$Ra = \frac{(9.81)(3 \times 10^{-4})(30)(0.01)^3}{(6.96 \times 10^{-7})(1.51 \times 10^{-7})}$$

$$Ra = 8.4 \times 10^5$$

9.4

$$mc \frac{dT}{dt} = -\bar{h}A(T - T_\infty)$$

$$\text{if } \bar{h} = C\Delta T^{1/4}$$

$$mc \frac{d\Delta T}{dt} = -AC\Delta T^{5/4}$$

$$\frac{d\Delta T}{dt} = -\frac{AC}{mc}\Delta T^{5/4} \longrightarrow \int \Delta T^{-5/4} d\Delta T = \int -\frac{AC}{mc} dt$$

integrating from $\Delta T_i = T_i - T_\infty$ at $t=0$ to $t=t_f$ at $\Delta T_f = T_f - T_\infty$

$$\Rightarrow -4\Delta T^{-1/4} \Big|_{T_i - T_\infty}^{T_f - T_\infty} = -\frac{AC}{mc} t_f \longrightarrow -\frac{AC}{mc} t_f = (T_f - T_\infty)^{-1/4} - (T_i - T_\infty)^{-1/4}$$

$$t_f = \frac{4mc}{AC} (T_f - T_\infty)^{-1/4} - (T_i - T_\infty)^{-1/4}$$